

Project Design Phase – II

Solution Requirements (Functional & Non-Functional)

Date	31 January 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the Hyderabad Metro Rail Digital Ticketing System built on the ServiceNow platform.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Service Portal Access	- Commuter accesses booking via ServiceNow Service Portal (/sp) - Commuter authenticates using ServiceNow credentials - Station Manager accesses Metro Database table via backend
FR-2	Catalog Item – Book A Metro Ticket	- Catalog item is created under Service Catalog > Services - Item is searchable and visible on the Service Portal - Item links to Metro Project flow on submission
FR-3	Variable Configuration	- Source station (Reference – Metro Station Details) - Destination station (Reference – Metro Station Details) - Number of passengers (String – validated numeric) - Type of journey (Choice: Single / Return) - Mode of payment (Choice: UPI / Card / Wallet / Cash / Others) - Enter payment mode (String – conditionally visible)
FR-4	Fare Auto-Calculation	- Catalog Client Script triggers on 'Type of Journey' change - Fare = No. of Passengers × Route Fare (per route mapping) - Single journey: populates 'Amount for Single Journey' - Return journey: populates 'Amount Including Return' (2× single)
FR-5	Field Validation	- OnChange Client Script validates 'No. of Passengers' - Alert shown if non-numeric value is entered - Mandatory fields enforced on form submission
FR-6	UI Policy – Conditional Display	- UI Policy triggers when Mode of Payment = 'Others' 'Enter Payment Mode' text field becomes visible - Field is hidden by default for all other payment modes
FR-7	QR Code Generation	- OnSubmit Client Script triggers on form submission - GlideModal opens with QR code generated from RITM sys_id - QR URL: <code>https://example.com/ticket?id=<sys_id></code> - Modal displays with title 'Scan Your QR Here'
FR-8	Flow Designer Automation (Metro Project)	- Trigger: Service Catalog – Requested Item submitted - Action 1: Get Catalog Variables (all booking variables) - Action 2: Create Record in Metro Database table - All catalog variables mapped to corresponding DB fields

FR-9	Data Storage – Metro Database Table	• Custom table stores all booking transaction records • Fields: Passenger Name, Source, Destination, Journey Type, No. of Passengers, Fare Amount, Payment Mode, Booking Date • Records accessible to authorized roles only (ACL-protected)
FR-10	Metro Station Details Table	• Custom table stores all metro station master data • Fields: Station Name, Station Code, Line, Zone, Address, Active • Referenced by catalog variables for Source and Destination • Populated with 10+ Hyderabad Metro stations
FR-11	Role-Based Access Control	• Default ACLs applied to Metro Database table • Commuters: can create (via flow) but cannot read table directly • Station Managers: read access to Metro Database • IT Admins: full CRUD access to all tables
FR-12	Service Portal – Metro Ticket Page	• Custom portal page created at /sp path • Metro QR Ticket widget embedded on the page • Page accessible to all authenticated portal users
FR-13	Business Rule – Generate Metro Ticket	• Business Rule fires on Metro Database record insert • Auto-generates ticket number / reference ID • Triggers email notification to requesting commuter

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution. These define the quality attributes, constraints, and operational characteristics of the system.

NFR No.	Non-Functional Requirement	Description	Target / Metric
NFR-1	Usability	The system must be intuitive and easy to use for all commuters.	Task completion rate ≥ 90% for non-technical users. - Form submission in under 2 minutes - Tooltip on every variable
NFR-2	Security	All data in the Metro Database must be protected via Role-Based Access Control (RBAC).	Access granted only to authorized personnel (admins, staff). - Zero unauthorized access in UAT - HTTPS enforced on all portal pages
NFR-3	Reliability	The system must reliably process every catalog submission and allow corresponding data to be stored in the Metro Database.	Availability ≥ 99.9% during peak hours. - Zero lost records in testing - Fallback error handling in client scripts
NFR-4	Performance	The catalog form must load within 3 seconds on the ServiceNow platform.	Portal page load ≤ 3 seconds for 95th percentile. - Fare auto-fill ≤ 1 second - QR modal display ≤ 2 seconds - Flow completion ≤ 5 seconds
NFR-5	Availability	The system runs on ServiceNow's cloud platform (SaaS), which provides 99.9%+ uptime.	ServiceNow SLA: 99.9%+ uptime. - PDI: available 24/7 during active use - No single point of failure (cloud-hosted)
NFR-6	Scalability	The ServiceNow platform scales automatically to handle increased load during peak hours.	Supports 1000+ concurrent users. - Concurrent multi-user booking supported - Station records expandable to 100+ stations - No code changes needed for scale-up
NFR-7	Maintainability	The system is built using standard ServiceNow components and follows best practices for development.	Change management via PDI. - Update sets exported per sprint - Code commented for readability
NFR-8	Portability	The solution is built on the ServiceNow platform and can be deployed to any ServiceNow instance.	Deployment via ServiceNow instance (PDI). - Platform-independent (cloud SaaS) - Compatible with ServiceNow Washington